

Entropy : it defines Randomness and disorders

$$Entropy = - \sum_{i=1}^N p_i \log_2 p_i \quad (9y, 5n)$$

Entropy of Root Node(9y , 5n)

$$Entropy = (-P_y) * (\log_2(P_y)) + (-P_n) * (\log_2(P_n))$$

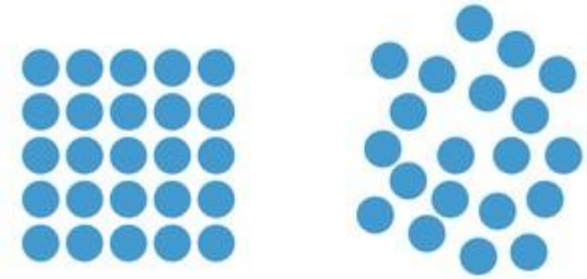
$$= \left(- \left(\frac{9}{14} \right) \right) * \left(\log_2 \left(\frac{9}{14} \right) \right) + \left(- \left(\frac{5}{14} \right) \right) * \left(\log_2 \left(\frac{5}{14} \right) \right)$$

$$= (-0.64) * (\log_2(0.64)) + (-0.35) * (\log_2(0.35))$$

$$= (-0.64) * (-0.64) + (-0.35) * (-1.49)$$

$$\mathbf{Entropy = 0.94}$$

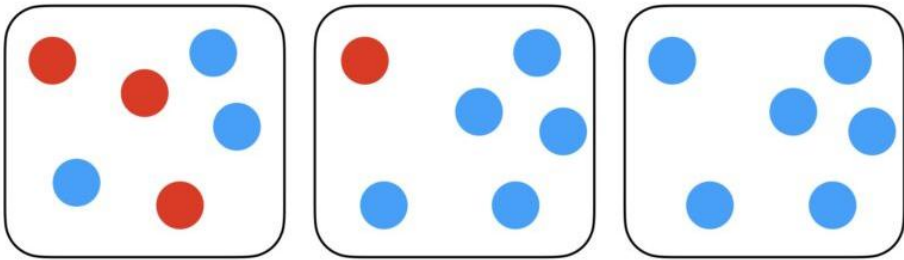
Entropy



Gini Impurity is a measure used in decision trees to determine how impure or mixed a dataset is. It represents the probability of incorrectly classifying a randomly chosen sample if it is labeled according to the class distribution in the dataset.

$$Gini = 1 - \sum_{i=1}^n (p_i)^2 \quad (9y, 5n)$$

$$Gini = 1 - \left[\left(\frac{y}{Total} \right)^2 + \left(\frac{n}{Total} \right)^2 \right]$$



GI of Root Node(9y , 5n)

$$Gini = 1 - \left[\left(\frac{9}{14} \right)^2 + \left(\frac{5}{14} \right)^2 \right]$$

$$= 1 - [(0.64)^2 + (0.36)^2]$$

$$= 1 - [0.41 + 0.13]$$

$$\mathbf{Gini = 0.46}$$